

Classification: Naïve Bayes' Classifier

- References

- http://www.cs.cmu.edu/~ninamf/courses/601sp15/slides/04_NBayes-1-26-2015.pptx.pdf
- <https://www3.cs.stonybrook.edu/~cse634/19Bayes2.pdf>
- http://www.cs.ucr.edu/~eamonn/CE/Bayesian%20Classification%20withInsect_examples.pdf

Let's learn classifiers by learning $P(Y|X)$

Consider $Y = \text{Wealth}$, $X = \langle \text{Gender}, \text{HoursWorked} \rangle$

gender	hours_worked	wealth		
Female	v0:40.5-	poor	0.253122	
		rich	0.0245895	
	v1:40.5+	poor	0.0421768	
		rich	0.0116293	
Male	v0:40.5-	poor	0.331313	
		rich	0.0971295	
	v1:40.5+	poor	0.134106	
		rich	0.105933	

Gender	HrsWorked	P(rich G,HW)	P(poor G,HW)
F	<40.5	.09	.91
F	>40.5	.21	.79
M	<40.5	.23	.77
M	>40.5	.38	.62

How many parameters must we estimate?

Suppose $X = \langle X_1, \dots, X_n \rangle$

where X_i and Y are boolean RV's

Gender	HrsWorked	$P(\text{rich} \mid G, HW)$	$P(\text{poor} \mid G, HW)$
F	<40.5	.09	.91
F	>40.5	.21	.79
M	<40.5	.23	.77
M	>40.5	.38	.62

To estimate $P(Y \mid X_1, X_2, \dots, X_n)$

If we have 30 boolean X_i 's: $P(Y \mid X_1, X_2, \dots, X_{30})$

How many parameters must we estimate?

Suppose $X = \langle X_1, \dots, X_n \rangle$

where X_i and Y are boolean RV's

$P(Y | X_1, \dots, X_n)$

Gender	HrsWorked	P(rich G, HW)	P(poor G, HW)
F	<40.5	.09	.91
F	>40.5	.21	.79
M	<40.5	.23	.77
M	>40.5	.38	.62

To estimate $P(Y | X_1, X_2, \dots, X_n)$

2^n

If we have 30 X_i 's instead of 2?

$2^{30} \sim 1 \text{ Billion}$

Bayes Rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Which is shorthand for:

$$(\forall i, j) P(Y = y_i | X = x_j) = \frac{P(X = x_j | Y = y_i) P(Y = y_i)}{P(X = x_j)}$$

Equivalently:

$$(\forall i, j) P(Y = y_i | X = x_j) = \frac{P(X = x_j | Y = y_i) P(Y = y_i)}{\sum_k P(X = x_j | Y = y_k) P(Y = y_k)}$$

Can we reduce params using Bayes Rule?

Suppose $X = \langle X_1, \dots, X_n \rangle$
where X_i and Y are boolean RV's

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

How many parameters to define $P(X_1, \dots, X_n | Y)$?

How many parameters to define $P(Y)$?

Can we reduce params using Bayes Rule?

Suppose $X = \langle X_1, \dots, X_n \rangle$

where X_i and Y are boolean RV's

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

how many params for $P(X_1 \dots X_n | Y)$ $(2^n - 1) \cdot 2$

how many for $P(Y) = 1$

Naïve Bayes

Naïve Bayes assumes

$$P(X_1 \dots X_n | Y) = \prod_i P(X_i | Y)$$

i.e., that X_i and X_j are conditionally independent given Y , for all $i \neq j$

Conditional Independence

Definition: X is conditionally independent of Y given Z , if the probability distribution governing X is independent of the value of Y , given the value of Z

$$(\forall i, j, k) P(X = x_i | Y = y_j, Z = z_k) = P(X = x_i | Z = z_k)$$

Which we often write

$$P(X|Y, Z) = P(X|Z)$$

E.g.,

$$P(\textit{Thunder} | \textit{Rain}, \textit{Lightning}) = P(\textit{Thunder} | \textit{Lightning})$$

Naïve Bayes uses assumption that the X_i are conditionally independent, given Y . E.g., $P(X_1|X_2, Y) = P(X_1|Y)$

Given this assumption, then:

$$\begin{aligned} P(X_1, X_2|Y) &= P(X_1|X_2, Y)P(X_2|Y) \\ &= P(X_1|Y)P(X_2|Y) \end{aligned}$$

in general:
$$P(X_1 \dots X_n|Y) = \prod_i P(X_i|Y)$$

How many parameters to describe $P(X_1 \dots X_n|Y)$? $P(Y)$?

- Without conditional indep assumption?
- With conditional indep assumption?

Naïve Bayes uses assumption that the X_i are conditionally independent, given Y

Given this assumption, then:

$$\begin{aligned} P(X_1, X_2|Y) &= P(X_1|X_2, Y)P(X_2|Y) \\ &= P(X_1|Y)P(X_2|Y) \end{aligned}$$

Chain rule
Cond. Indep.

in general: $P(X_1 \dots X_n|Y) = \prod_i P(X_i|Y)$

How many parameters to describe $P(X_1 \dots X_n|Y)$? $P(Y)$?

- Without conditional indep assumption? $2(2^n - 1) + 1$
- With conditional indep assumption? $2^n + 1$

Naïve Bayes in a Nutshell

Bayes rule:

$$P(Y = y_k | X_1 \dots X_n) = \frac{P(Y = y_k) P(X_1 \dots X_n | Y = y_k)}{\sum_j P(Y = y_j) P(X_1 \dots X_n | Y = y_j)}$$

Assuming conditional independence among X_i 's:

$$P(Y = y_k | X_1 \dots X_n) = \frac{P(Y = y_k) \prod_i P(X_i | Y = y_k)}{\sum_j P(Y = y_j) \prod_i P(X_i | Y = y_j)}$$

So, to pick most probable Y for $X^{new} = \langle X_1, \dots, X_n \rangle$

$$Y^{new} \leftarrow \arg \max_{y_k} P(Y = y_k) \prod_i P(X_i^{new} | Y = y_k)$$

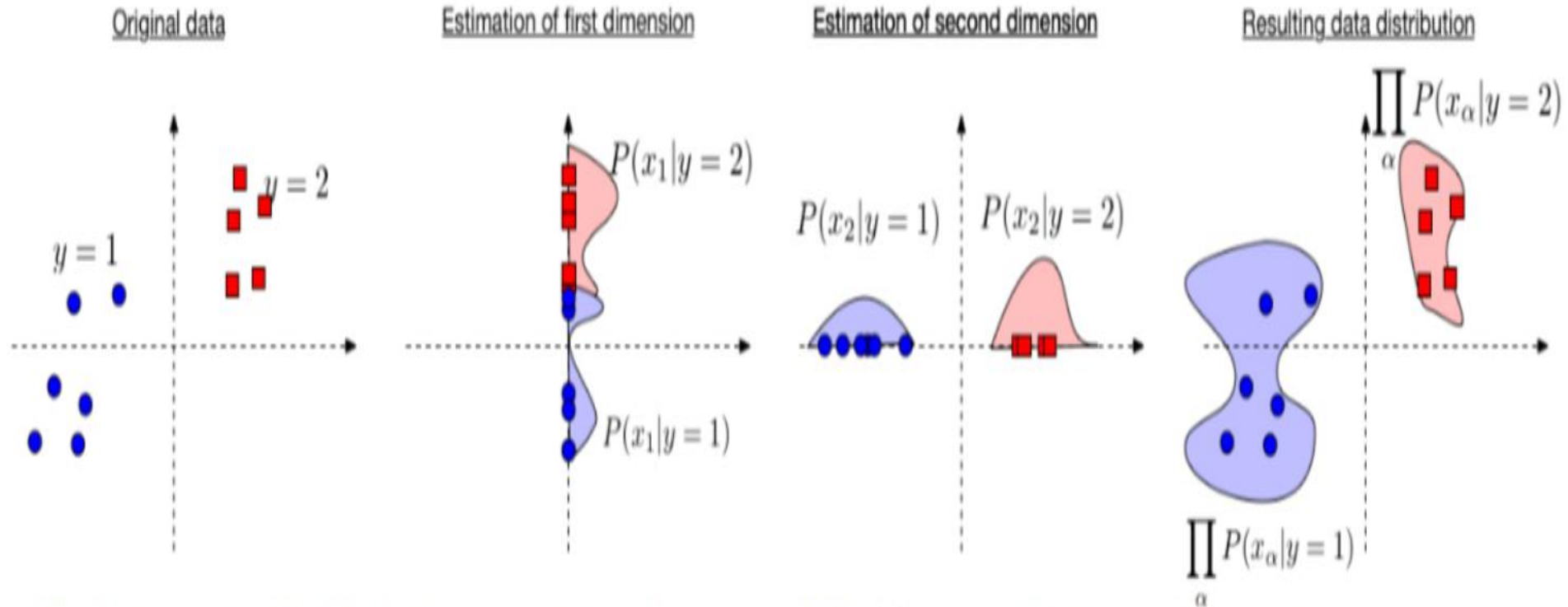


Illustration behind the Naive Bayes algorithm. We estimate $P(x_\alpha|y)$ independently in each dimension (middle two images) and then obtain an estimate of the full data distribution by assuming conditional independence $P(\mathbf{x}|y) = \prod_{\alpha} P(x_\alpha|y)$ (very right image).

<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote05.html>

Naïve Bayes Classifier

- ▶ A naïve Bayes (NB) classifier is a simple probabilistic classifier based on: (a) Bayes theorem, (b) strong (naïve) independence assumptions, and (c) independent feature models.
- ▶ It is also an important mining classifier for pattern classification and applied in many real world classification problems because of its high classification performance.
- ▶ A NB classifier can easily handle missing attribute values by simply omitting the corresponding probabilities for those attributes when calculating the likelihood of membership for each class.
- ▶ The NB classifier also requires the class conditional independence, i.e. the effect of an attribute on a given class is independent of those of other attributes.

Dataset

- ▶ Given a training dataset, $D = \{X_1, X_2, \dots, X_n\}$, each data record is represented as, $X_i = \{x_1, x_2, \dots, x_n\}$.
- ▶ D contains the following attributes $\{A_1, A_2, \dots, A_n\}$ and each attribute A_i contains the following attribute values $\{A_{i1}, A_{i2}, \dots, A_{ih}\}$.
- ▶ The attribute values can be discrete or continuous.
- ▶ D also contains a set of classes $C = \{C_1, C_2, \dots, C_m\}$. Each training instance, $X \in D$, has a particular class label C_i .
- ▶ For a test instance, X , the classifier will predict that X belongs to the class with the highest posterior probability, conditioned on X .

NB classifier

The NB classifier predicts that the instance X belongs to the class C_i , if and only if $P(C_i|X) > P(C_j|X)$ for $1 \leq j \leq m, j \neq i$. The class C_i for which $P(C_i|X)$ is maximized is called the Maximum Posteriori Hypothesis.

$$P(C_i|X) = \frac{P(X|C_i)P(C_i)}{P(X)} \quad (9)$$

In Bayes theorem shown in Eq. 9, as $P(X)$ is a constant for all classes, only $P(X|C_i)P(C_i)$ needs to be maximized. If the class prior probabilities are not known, then it is commonly assumed that the classes are equally likely, that is, $P(C_1) = P(C_2) = \dots = P(C_m)$, and therefore maximize $P(X|C_i)$. Otherwise, maximize $P(X|C_i)P(C_i)$. The class prior probabilities are calculated by $P(C_i) = |C_{i,D}|/|D|$, where $|C_{i,D}|$ is the number of training instances belonging to the class C_i in D .

NB classifier (con.)

To compute $P(X|C_i)$ in a dataset with many attributes is extremely computationally expensive. Thus, the naïve assumption of class conditional independence is made in order to reduce computation in evaluating $P(X|C_i)$. The attributes are conditionally independent of one another, given the class label of the instance. Thus, Eq. 10 and Eq. 11 are used to produce $P(X|C_i)$.

$$P(X|C_i) = \prod_{k=1}^n P(x_k|C_i) \quad (10)$$

$$P(X|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times \cdots \times P(x_n|C_i) \quad (11)$$

In Eq. 10, x_k refers to the value of attribute A_k for instance X .

Therefore, these probabilities $P(x_1|C_i), P(x_2|C_i), \cdots, P(x_n|C_i)$ can be easily estimated from the training instances.

NB classifier (con.)

Moreover, the attributes in training datasets can be categorical or continuous-valued. If the attribute value, A_k , is categorical, then $P(x_k|C_i)$ is the number of instances in the class $C_i \in D$ with the value x_k for A_k , divided by $|C_{i,D}|$, i.e. the number of instances belonging to the class $C_i \in D$. If A_k is a continuous-valued attribute, then A_k is typically assumed to have a Gaussian distribution with a mean μ and standard deviation σ , defined respectively by the following two equations:

$$P(x_k|C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i}) \quad (12)$$

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (13)$$

In Eq. 12, μ_{C_i} is the mean and σ_{C_i} is the standard deviation of the values of the attribute A_k for all training instances in the class C_i . Now we can bring these two quantities to Eq. 13, together with x_k , in order to estimate $P(x_k|C_i)$.

Algorithm 1 Naïve Bayes classifier

Input: $D = \{x_1, x_2, \dots, x_n\}$ // Training data.

Output: A naïve Bayes Model.

Method:

```
1: for each class,  $C_i \in D$ , do
2:   Find the prior probabilities,  $P(C_i)$ .
3: end for
4: for each attribute,  $A_i \in D$ , do
5:   for each attribute value,  $A_{ij} \in A_i$ , do
6:     Find the class conditional probabilities,  $P(A_{ij}|C_i)$ .
7:   end for
8: end for
9: for each instance,  $x_i \in D$ , do
10:  Find the posterior probability,  $P(C_i|x_i)$ ;
11: end for
```

NB classifier (con.)

To predict the class label of instance X , $P(X|C_i)P(C_i)$ is evaluated for each class $C_i \in D$. The NB classifier predicts that the class label of instance X is the class C_i , if and only if

$$P(X|C_i)P(C_i) > P(X|C_j)P(C_j) \quad (14)$$

In Eq. 14, $1 \leq j \leq m$ and $j \neq i$. That is the predicted class label is the class C_i for which $P(X|C_i)P(C_i)$ is the maximum probability. Algorithm 1 outlines the naïve Bayes classifier algorithm.

Laplace Correction

- ▶ A zero probability cancels the effects of all the other (posteriori) probabilities (on C_i) involved in the product.
- ▶ We can assume that our training database, D , is so large that adding one to each count that we need would only make a negligible difference in the estimated probability value, yet would conveniently avoid the case of probability values of zero.
- ▶ This technique estimation is known as the **Laplacian correction** or **Laplace estimator**, named after Pierre Laplace, a French mathematician who lived from 1749 to 1827.

unsmoothed

$$p(X_1 = T \mid Y = \text{spam}) = \frac{\text{freq. of } T \text{ in spam data}}{\text{total \# of spam instances}}$$

smoothed

$$p(X_1 = T \mid Y = \text{spam}) = \frac{\text{freq. of } T \text{ in spam} + 1}{\text{total \# of spam instances} + v}$$

Classification Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- A married person with income 120K did not refund the loan previously
- *Can we trust him?*

Naïve Bayes Classifier

- Assume independence among attributes A_i when class is given:
 - $P(A_1, A_2, \dots, A_n | C) = P(A_1 | C_j) P(A_2 | C_j) \dots P(A_n | C_j)$
 - Can estimate $P(A_i | C_j)$ for all A_i and C_j .
 - *the new pattern* is classified to C_j if $P(C_j) \prod P(A_i | C_j)$ is maximal.

How to Estimate Probabilities from Data?

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Class: $P(C) = N_c/N$
 - e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$

- For discrete attributes:

$$P(A_i | C_k) = |A_{ik}| / N_c$$

- where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class C_k
- Examples:

$$P(\text{Status}=\text{Married}|\text{No}) = 4/7$$
$$P(\text{Refund}=\text{Yes}|\text{Yes})=0$$

How to Estimate Probabilities from Data?

- For continuous attributes:
 - Discretize the range into bins
 - one ordinal attribute per bin
 - violates independence assumption
 - Two-way split: $(A < v)$ or $(A > v)$
 - choose only one of the two splits as new attribute
 - Probability density estimation:
 - Assume attribute follows a normal distribution
 - Use data to estimate parameters of distribution (e.g., mean and standard deviation)
 - Once probability distribution is known, can use it to estimate the conditional probability $P(A_i | c)$

How to Estimate Probabilities from Data?

<i>Tid</i>	<u>Refund</u>	<u>Marital Status</u>	<u>Taxable Income</u>	<u>Evade</u>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Normal distribution:

$$P(A_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

- One for each (A_i, c_i) pair
- For (Income, Class=No):
 - If Class=No
 - sample mean = 110K
 - sample variance = 2975

How to Estimate Probabilities from Data?

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
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- Normal distribution:

$$P(A_i | c_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

- One for each (A_i, c_i) pair
- For (Income, Class=No):
 - If Class=No
 - sample mean = 110K
 - sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

Example of Naïve Bayes Classifier

Given a Test Record: $X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$

naive Bayes Classifier:

$$P(\text{Refund}=\text{Yes}|\text{No}) = 3/7$$

$$P(\text{Refund}=\text{No}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes}) = 0$$

$$P(\text{Refund}=\text{No}|\text{Yes}) = 1$$

$$P(\text{Marital Status}=\text{Single}|\text{No}) = 2/7$$

$$P(\text{Marital Status}=\text{Divorced}|\text{No}) = 1/7$$

$$P(\text{Marital Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Marital Status}=\text{Single}|\text{Yes}) = 2/7$$

$$P(\text{Marital Status}=\text{Divorced}|\text{Yes}) = 1/7$$

$$P(\text{Marital Status}=\text{Married}|\text{Yes}) = 0$$

For taxable income:

If class=No: sample mean=110
 sample variance=2975

If class=Yes: sample mean=90
 sample variance=25

- $P(X|\text{Class}=\text{No}) = P(\text{Refund}=\text{No}|\text{Class}=\text{No})$
 $\times P(\text{Married}|\text{Class}=\text{No})$
 $\times P(\text{Income}=120\text{K}|\text{Class}=\text{No})$
 $= 4/7 \times 4/7 \times 0.0072 = 0.0024$
- $P(X|\text{Class}=\text{Yes}) = P(\text{Refund}=\text{No}|\text{Class}=\text{Yes})$
 $\times P(\text{Married}|\text{Class}=\text{Yes})$
 $\times P(\text{Income}=120\text{K}|\text{Class}=\text{Yes})$
 $= 1 \times 0 \times 1.2 \times 10^{-9} = 0$

Since $P(X|\text{No})P(\text{No}) > P(X|\text{Yes})P(\text{Yes})$

Therefore $P(\text{No}|X) > P(\text{Yes}|X)$
 $\Rightarrow \text{Class} = \text{No}$

Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

A: attributes

M: mammals

N: non-mammals

$$P(A|M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A|N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A|M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A|N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

$$P(A|M)P(M) >$$

$$P(A|N)P(N)$$

=> Mammals

Simple demonstration of Spam detection

- ❑ Reviews{Non-spam(+), Spam(-)}
- ❑ Represent each review by vector of words
- ❑ Learning: Use training examples to estimate
 - ❑ $P(+)$
 - ❑ $P(-)$
 - ❑ $P(\text{rev}|+)$
 - ❑ $P(\text{rev}|-)$
- ❑ Naïve Bayes condition independence assumption.

Simple demonstration of Spam detection

- ❑ Set of reviews and their classification

Doc	Review	Class
1	I loved this hotel	+
2	I hated this hotel	-
3	A great hotel, good hotel	+
4	Bad service	-
5	Great service, a good hotel	+

- ❑ 10 unique words

<I, loved, this, hotel, hated, a, great, bad, service, good>

Simple demonstration of Spam detection

- ❑ Converting the reviews into feature sets.

Doc	I	loved	this	Hotel	Hated	A	Great	Bad	Service	Good	Class
1	1	1	1	1							+
2	1		1	1	1						-
3				2		1	1			1	+
4								1	1		-
5				1		1	1		1	1	+

Simple demonstration of Spam detection

- ❑ Reviews with positive outcomes.

Doc	I	loved	this	Hotel	Hated	A	Great	Bad	Service	Good	Class
1	1	1	1	1							+
3				2		1	1			1	+
5				1		1	1		1	1	+

- ❑ $P(+)= 3/5= 0.6$
- ❑ Compute: $p(I|+)$; $p(\text{loved}|+)$; $p(\text{this}|+)$; $p(\text{hotel}|+)$; $p(a|+)$; $p(\text{great}|+)$; $p(\text{service}|+)$; $p(\text{good}|+)$; $p(\text{hated}|+)$; $p(\text{bad}|+)$.

Simple demonstration of Spam detection

- $P(+)=3/5=0.6;$ $p(w_k|+)= (n_k+1)/(n+|\text{vocabulary}|)$
 - $p(l|+): (1+1)/(14+10)= 0.0833;$
 - $p(liked|+): (1+1)/(14+10)= 0.0833;$
 - $p(this|+): (1+1)/(14+10)= 0.0833;$
 - $p(hotel|+): (5+1)/(14+10)= 0.2083;$
 - $p(a|+): (2+1)/(14+10)= 0.125;$
 - $p(great|+): (2+1)/(14+10)= 0.125;$
 - $p(service|+): (1+1)/(14+10)= 0.0833;$
 - $p(good|+): (2+1)/(14+10)= 0.125;$
 - $p(hated|+): (0+1)/(14+10)= 0.0417;$
 - $p(bad|+): (0+1)/(14+10)= 0.0417;$

Simple demonstration of Spam detection

- ❑ Reviews with negative outcomes.

Doc	I	loved	this	Hotel	Hated	A	Great	Bad	Service	Good	Class
2	1		1	1	1						-
4								1	1		-

- ❑ $P(-) = 2/5 = 0.4$
- ❑ Compute: $p(I|-)$; $p(\text{loved}|-)$; $p(\text{this}|-)$; $p(\text{hotel}|-)$; $p(a|-)$; $p(\text{great}|-)$; $p(\text{service}|-)$; $p(\text{good}|-)$; $p(\text{hated}|-)$; $p(\text{bad}|-)$.

Simple demonstration of Spam detection

□ $P(-) = 2/5 = 0.4;$ $p(w_k | -) = (n_k + 1) / (n + |\text{vocabulary}|)$

$p(I | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{this} | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{hotel} | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{hated} | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{bad} | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{service} | -) = (1 + 1) / (6 + 10) = 0.125;$

$p(\text{loved} | -) = (0 + 1) / (6 + 10) = 0.0625;$

$p(a | -) = (0 + 1) / (6 + 10) = 0.0625;$

$p(\text{great} | -) = (0 + 1) / (6 + 10) = 0.0625;$

$p(\text{good} | -) = (0 + 1) / (6 + 10) = 0.0625;$

Simple demonstration of Spam detection

- ❑ Now training of our classifier is done,
- ❑ Lets test the classifier with a test Review:

“I hated this bad service”

If $v_j = +$; $P(+)P(I | +)P(hated | +)P(this | +)P(bad | +)P(service | +) =$
 6.03×10^{-7}

If $v_j = -$; $P(-)P(I | -)P(hated | -)P(this | -)P(bad | -)P(service | -) =$
 1.22×10^{-5}



Spam

A Practice Example

Example 8.4

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

Data instance

X = (age ≤ 30,

Income = medium,

Student = yes

Credit_rating = fair)

age	income	student	credit_rating	com
≤30	high	no	fair	no
≤30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

A Practice Example

- $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$
 $P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$
- Compute $P(X|C_i)$ for each class
 $P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$
 $P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$
 $P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$
 $P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$
 $P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$
 $P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$
 $P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$
 $P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$
- $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$
 $P(X|C_i) : P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$
 $P(X|\text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$
 $P(X|C_i) * P(C_i) : P(X|\text{buys_computer} = \text{"yes"}) * P(\text{buys_computer} = \text{"yes"}) = 0.028$
 $P(X|\text{buys_computer} = \text{"no"}) * P(\text{buys_computer} = \text{"no"}) = 0.007$

Therefore, X belongs to class ("buys_computer = yes")

